Zuri Ye

+1 (424) 781-2931 | zuriye45@gmail.com | linkedin.com/in/zuriye/ | github.com/zuri2102 | https://zuri2102.github.io

EDUCATION

CARNEGIE MELLON UNIVERSITY — Cumulative GPA: 3.91/4.0

Major in **Electrical and Computer Engineering**, Additional Major in **Robotics** — Class of 2029

Relevant Coursework: Computer Systems; Computer Vision; Data Structures; Analog Circuits; Digital Logic Design; Linear Algebra; Discrete; Calc III

TECHNICAL EXPERIENCE

RESILIENT INTELLIGENT SYSTEMS LAB (RISLAB) | GITHUB — November 2025 - Present

Undergraduate Researcher

- Researching real-time SLAM optimization for high-speed UAVs to mitigate motion blur and large inter-frame displacement
- Implementing Gaussian Splatting monocular SLAM from VINGS-Mono—VIO, optical flow, IMU preintegration, sensor fusion, loop closure, pose estimation, Lie Group Transformations—to leverage 3D mapping/SLAM pipelines
- Optimizing feature association across wide spatial gaps with factor graphs, GTSAM, and GRUs/CNNs in PyTorch/CUDA

TARTAN AUV — September 2025 - Present

Electrical Subteam Member

- Designing custom PCBs in Altium Designer for power distribution and servo breakout on an autonomous underwater vehicle
- Assembling PCBs via SMD, through-hole, and reflow while supporting electrical integration and hardware across the vehicle

HONEY HAVEN | GITHUB | GAME — January 2026 - May 2026

Project Co-Lead, Programming Lead, Art Lead

- Led full-lifecycle development and Itch.io release of a visual novel, winning 1st Place at a game expo with 200+ plays
- Architected a modular GDScript pipeline separating logic, UI, and minigames so subteams could work in parallel safely
- Built a standalone Twine-to-JSON content layer dynamically parsing scripts into UI, character state, and scene cues, letting writers iterate on narrative content independent of game code, with zero recompilation or engineering bottleneck

USS HORNET MUSEUM — July 2023 - August 2025

Aircraft Restoration Group Volunteer

- Restored Vietnam aircraft (T-28B, S-3, F-4), building practical power-tool and shop-safety skills

MIT LL BEAVER WORKS SUMMER INSTITUTE | GITHUB — July 2024 - August 2024

Unmanned Air Vehicles Student and Software Communications Lead

- Engineered a ROS2-based autonomous quadcopter, leading and designing a node architecture where concurrent processes for control, and perception (designed and written using Python and OpenCV) communicated over pub/sub topics in real time

PROJECTS

HUAWEI U2800A CANDYBAR PHONE-IPOD — In Progress

- Designing a custom PCB MP3 player from scratch in KiCAD to fit inside a repurposed Huawei U2800A candybar shell, constraining component placement, trace routing, and connector orientation to a fixed, non-negotiable enclosure footprint
- Architected an ESP32-WROOM-32E board, multiplexing SPI/I2C/I2S across TFT, microSD, and audio under pin/bus constraints; built USB-C LiPo soft-shutdown/reset power handling at the firmware-hardware boundary
- Implementing Wi-Fi playlist sync and offline microSD storage, with a planned Spotify-sync companion app, balancing onboard flash/RAM limits against streaming vs. local playback tradeoffs

SYSTEMS COURSEWORK — May 2026 - July 2026

- Built a dynamic allocator, multi-level cache simulator, Unix shell, and concurrent network proxy in C, applying virtual memory, scheduling, and signal handling
- Debugged at the assembly/register level with GDB/objdump, developing fluency in x86-64 low-level execution

SIR JESTER | GITHUB | GAME — February 2026

- 15kB zipped game designed with minimized footprint through byte-level tradeoffs: baked-in pixel art and 8-bit music at runtime instead of storing binary assets, shortened/mangled variable names, and inlined a single-file HTML/CSS/JS architecture

ADDITIONAL SKILLS

Languages & Tools: C, Python, Java, ASM, GDB, R, MATLAB, OpenCV, ROS2, Linux, Bash, Git, GitHub
CAD & Fabrication: Altium Designer, KiCAD, OnShape, PCB assembly & soldering, 3D printing, microcontrollers
Technical Domains: computer vision, sensor fusion, embedded systems, machine learning, prompt engineering